

Brain2Text

Neural Decoding — Experiment Design

Baseline WER 0.3673 · Target WER < 0.10 · RTX 4050 6 GB (local) + A100 40 GB (cloud)

5

Tracks

16

Experiments

12

Run Locally

4

Cloud Only

~20 min

Toy Run Time

Experiment Tracks

Five orthogonal investigation axes — each independently testable, combined for maximum WER reduction



Track A

Pretraining Modality

Audio vs image/video vs text LLM — why does modality of pretraining affect neural decoding quality?

4 exp



Track B

Encoder Architecture

ConformerXL, HRM, MambaPOSSM, MoE, ZenBrain — drop-in BIT_Transformer replacements.

5 exp



Track C

LLM Decoder Variants

Qwen2-Audio-7B, Phi4-Multimodal, Whisper-Qwen — audio-pretrained LLM backbone ablations.

3 exp



Track D

Loss Function Design

CTC weight annealing, contrastive weight, topographic loss, label smoothing.

4 exp



Track E

Projector Architecture

MLP depth variants, QFormer cross-attention projector, patch-size token-count grid.

3 exp

Baseline Architecture · Toy Training Protocol

Current E2E Pipeline



Best WER 0.3673 · Epoch 65 / 150 · LR = 6.25e-6

Inference fix applied: nucleus sampling top_p=0.9 · temperature=0.7 · max_new_tokens=25

Toy Training Protocol

	Full Run	Toy Run
Epochs	150	20
Batches / epoch	1,369	200
Total time	~15 hours	~20 min
Data fraction	100%	15%
Val interval	5 epochs	2 epochs
Batch size	8	4

Promotion Rule (WER slope improvement vs baseline)

≥ 5% rel. → Full A100 run immediately

3 – 5% rel. → Combine with best other change, re-run toy

1 – 3% rel. → Only pursue if compute is idle

< 1% rel. → Discard

TRACK A

Pretraining Modality Study

Audio vs image/video vs text-only LLMs — which prior helps neural decoding, and why?

Track A · Four Competing Hypotheses

Each hypothesis implies a different experimental test — and a different architectural fix.

H1 — Phoneme Prior

Audio pretraining creates richer sub-word phoneme representations. Neural speech encodes phonemes — audio LLMs understand phoneme-to-word mappings better than text LLMs.

Experiment A3: Phoneme probing accuracy on frozen LLM hidden states

H2 — Spoken Language Prior

Spoken text statistics differ from written text: shorter sentences, simpler syntax, more repetition. Audio-pretrained LLMs are better calibrated for spoken sentence distributions.

Experiment A2: Perplexity ratio (spoken / written corpus) across models

H3 — Embedding Geometry

The token embedding space of audio-pretrained LLMs is geometrically closer to what the BIT projector produces. The projector bridges a shorter semantic distance.

Experiment A1: CKA alignment between neural projections and LLM embeddings

H4 — Temporal Structure

Pretraining on video (also a temporal sequence modality) may help similarly to audio — suggesting the benefit is about temporal structure generally, not speech specifically.

Experiment A4: E2E WER with Qwen2.5-VL-3B (video) vs Qwen2-Audio-7B (audio)

Track A · Experiments A1 – A4

A1

LOCAL

CKA Embedding Alignment

Method

Compute linear CKA between BIT encoder output and each LLM's embedding space over the validation set. No training required — pure geometry measurement.

Models

5 LLMs: text-1.5B, text-7B, vision-3B, audio-7B, phi4-MM

Metric

CKA score — higher = better alignment between spaces

30 min

A2

LOCAL

Spoken Language Perplexity

Method

Compute perplexity of each frozen LLM on BCI competition sentences (spoken) vs Wikipedia excerpts (written). Tests whether audio pretraining narrows the spoken/written gap.

Models

Qwen2.5-1.5B · Qwen2.5-7B · Qwen2-Audio-7B

Metric

PPL_spoken / PPL_written — lower ratio = better spoken prior

15 min

A3

LOCAL

Phoneme Probing

Method

Freeze LLM. Train Linear(llm_dim → 42) probe on hidden states to predict phoneme class from ground-truth text. Tests whether phoneme information is encoded in LLM representations.

Models

3 frozen LLMs compared with identical probe and data

Metric

Phoneme top-1 classification accuracy

45 min

A4

CLOUD

Full E2E WER Comparison

Method

Train each decoder (text / audio / vision / Whisper-Qwen) on toy data for 20 epochs. Compare WER learning slope. Whisper-Qwen (C3) can be run locally as A4 proxy.

Models

4 LLM decoders × 20 toy epochs

Metric

WER at epoch 10 + slope (epoch 2 → 20)

~3 hr cloud

TRACK B

Encoder

Architectures

Five drop-in replacements for BIT_Transformer — all output (B, T_patch, 384)

Track B · Architecture Comparison

Architecture	Key Innovation	Extra Params	Session Specific?	Local
BIT (baseline)	RoPE Transformer · 7 layers · MAE SSL pretrained · per-session ModuleDict read-in layers	—	✓	✓
ConformerXL	JitterCorrectionPrenet: dilated conv (d=1,2,4) + BiGRU corrects spike-timing jitter before patching. Macaron FFN + ConvModule	+3M	✗	✓
HRM	Dual-timescale GRU: L-module (20ms bins, GRUCell) + H-module (100ms patches). Fixed-point DEQ solver — O(1) memory per patch	−10M	✗	✓
MambaPOSSM	Cross-attention IndividualSpikeTokenizer replaces linear patch embedding. SSM (Mamba / GRU fallback) with input-dependent state selection	+1M	✗	✓
MoE Encoder	6 specific experts (top-2 routing) + 2 shared experts replace FFN in all 7 TransformerBlocks. TopK router with load-balance auxiliary loss	+40M	✗	✓
ZenBrain	DualMemoryBlock: cross-attention over episodic buffer (M=3000 patches). Buffer fills from high-confidence trials — zero recalibration at test	+2M	✗	✓

All encoders: `forward(x, session_id=None, mask_patches=None, neural_lengths=None) → (B, T_patch, 384)` · No SSL pretraining checkpoint for non-BIT encoders — train from scratch for fair comparison.

Track B · Architecture Deep-Dives

ConformerXL

Why Spike-timing jitter: neurons fire within $\pm 10\text{ms}$ of their 'true' time. BIT embeds raw bins, propagating jitter into patch representations.

How JitterCorrectionPrenet: 3 dilated conv branches ($d=1,2,4$) + BiGRU, runs before patching. Macaron blocks: $\text{FFN} \times 0.5 \rightarrow \text{MHSA} \rightarrow \text{DepthwiseConv} \rightarrow \text{FFN} \times 0.5$.

MambaPOSSM (SSM)

Why At any 20ms bin, only ~ 50 of 512 channels fire. Linear patch embedding weights all channels equally. Attention-based tokenization selects informative channels.

How IndividualSpikeTokenizer: 64 learned queries cross-attend to 4 KV frames $\rightarrow$ sum-pool $\rightarrow$ 1 token. SSMBlock: Mamba (or GRU fallback) with input-dependent state transitions.

ZenBrain (Episodic Memory)

Why BCI electrode signals drift across trials. High-quality decodings from earlier in a session should help interpret noisier later trials — like in-context learning.

How EpisodicBuffer: ring buffer $M=3000 \times 384$. DualMemoryBlock adds cross-attn over buffer. `use_memory=False` during training; buffer fills from CTC-confident patches at inference.

HRM (Hierarchical Recurrent)

Why Neural speech has two timescales: individual spikes (20ms) and phoneme patterns (100ms). A single transformer treats all patches uniformly.

How L-module (GRUCell at 20ms, gated by H context) + H-module (GRUCell over patch summaries). DEQ fixed-point solver per patch — $O(1)$ memory, no cross-patch BPTT.

MoE Encoder

Why Neural data is multi-domain: phonemes, prosody, motor commands, noise. A single FFN processes all token types identically — experts could specialize.

How SSMoEBlock: 6 routed experts (top-2 per token via TopK router) + 2 always-active shared experts. Load-balance auxiliary loss ($\lambda=0.01$) prevents expert collapse.

Track C · LLM Decoder Variants

BIT encoder is held constant. Only the LLM decoder changes.

C1

CLOUD

Qwen2-Audio-7B -Instruct

LLM dim	3584
Parameters	7B · ~7 GB at 4-bit NF4
LoRA targets	q, k, v, o, gate, up, down

Pipeline

Loads as AutoModelForCausalLM — language backbone shaped by joint audio-text training. BIT embeddings injected via inputs_embeds, bypassing Qwen2-Audio's built-in Whisper encoder.

Closest equivalent to paper's Aero1-Audio. Most principled audio pretraining test.

C2

CLOUD

Phi-4-Multimodal -Instruct

LLM dim	3072
Parameters	5.6B · ~6 GB at 4-bit NF4
LoRA targets	qkv_proj, o_proj, gate_up_proj, down_proj

Pipeline

Speech + video + image pretrained (Azure Speech data). Uses fused QKV — LoRA targets qkv_proj covering all three heads in one matrix. Both temporal and acoustic priors.

Smaller than C1 but trained on both temporal (video) and acoustic (speech) modalities.

C3

LOCAL

Whisper-Large-v3 + Qwen2.5-1.5B

LLM dim	1280 → 1536
Parameters	615M (frozen) + 1.5B + LoRA
LoRA targets	Qwen2.5 only (7 standard targets)

Pipeline

Frozen Whisper encoder acts as speech manifold normalizer: BIT output → Linear(384, 1280)+LN → frozen Whisper encoder → Linear(1280, 1536) → Qwen2.5 + LoRA.

Runs on RTX 4050 (~4.1 GB). Most principled test of the embedding alignment hypothesis (H3).

Track D · Loss Design Track E · Projector Architecture

$$\text{Loss} = \text{CE} + 0.3 \times \text{CTC} + 1.0 \times \text{Contrastive}$$

D1 CTC Annealing

0.3→0.0 linearly over 75 epochs. CTC is most valuable early; fixed weight may constrain late E2E specialization. Variants: fixed / linear / step / zero.

D2 Contrastive Weight

Test weights 0.0 / 0.1 / 1.0 / 2.0. If weight=0 gives better WER, the contrastive loss conflicts with CE and should be removed or rescheduled.

D3 Topographic Loss

Forces FFN weights to form cortical-like spatial maps. Pure regularizer — zero new params. Targets FFN first-linear in all TransformerBlocks. Weight $\in \{0.001, 0.01, 0.1\}$.

D4 Label Smoothing

$\epsilon \in \{0, 0.05, 0.10, 0.20\}$. With ~1000 BCI samples the model memorizes sequences. Smoothing prevents overconfidence on training sentences.

E1–E3 Projector

E1 MLP Variants

3-layer MLP (current) vs 5-layer vs Gated MLP with multiplicative interaction. Tests whether projector depth limits semantic bridge quality.

E2 Q-Former Cross-Attention Projector

N learned query tokens cross-attend over the full neural sequence → project to LLM dim. Fixed-length output ($N_Q \in \{16, 32, 64\}$) regardless of neural sequence length. No padding waste in LLM attention. Highest-impact projector change expected.

E3 Patch-size × Token-count Grid

2×3 grid: patch_size $\in \{4, 5, 8\} \times N_Q \in \{16, 32, 64\}$. Paper uses patch_size=5.

Local Feasibility · 5-Phase Roadmap

Hardware Requirements

A1–A3 CKA · Perplexity · Probe	~4 GB
B1 ConformerXL + Qwen1.5B	~3 GB
B2 HRM + Qwen1.5B	~2.5 GB
B3 MambaPOSSM + Qwen1.5B	~3 GB
B4 MoE Encoder + Qwen1.5B	~3.5 GB
B5 ZenBrain + Qwen1.5B	~3 GB
C1 Qwen2-Audio-7B decoder	~8 GB
C2 Phi4-Multimodal decoder	~7 GB
C3 WhisperQwen (A4 proxy)	~4.1 GB
D1–D4 Loss ablations	~3 GB
E1–E3 Projector variants	~3 GB

5-Phase Experiment Roadmap

01	Analysis — No Training	2 hr
A1 CKA · A2 Perplexity · A3 Phoneme Probe (local, inference only)		
02	Toy Encoder Ablations	3 hr
B1–B5 on 15% toy data, 20 epochs each. BIT from scratch as baseline.		
03	Loss + Projector Ablations	5 hr
D1b, D2a/d, D3b/c, E2b (Q-Former), E3 grid. All local toy runs.		
04	Cloud — Top 2 Architectures	6 hr
Best encoder × best projector/loss combo. Full 80-epoch A100 runs.		
05	Audio LLM Experiments	3 hr
C1 Qwen2-Audio · A4 vision comparison. Full-data A100 training.		

When to promote a toy result to full cloud training

≥ 5% relative WER slope improvement

→ Run full 80-epoch A100 experiment immediately

Slope metric = $(\text{WER_epoch2} - \text{WER_epoch20}) / 18$ compared against BIT-from-scratch on identical toy data

3 – 5% relative

→ Combine with best complementary change, re-run toy

Slope metric = $(\text{WER_epoch2} - \text{WER_epoch20}) / 18$ compared against BIT-from-scratch on identical toy data

1 – 3% relative

→ Pursue only if cloud compute is currently idle

Slope metric = $(\text{WER_epoch2} - \text{WER_epoch20}) / 18$ compared against BIT-from-scratch on identical toy data

< 1% relative

→ Discard — within noise level

Slope metric = $(\text{WER_epoch2} - \text{WER_epoch20}) / 18$ compared against BIT-from-scratch on identical toy data